25 Manufacturing Execution Systems & Business Integration

By Dennis Brandl

Topic Highlights

MES Integration with Business Planning and Logistics
Level 3 Equipment Hierarchy
MES and Production Operations Management
Detailed Production Scheduling
- *Production Dispatching*
- *Production Execution Management*
- *Production Data Collection*
- *Production Tracking*
- *Production Resource Management*
- *Product Definition Management*
- *Production Performance Analysis*

Other Manufacturing Activities
Level 3-4 Boundary

25.1 Introduction

Automation does not end with equipment control; it also includes higher levels of control that manage personnel, equipment, and materials across production areas. Effectiveness in manufacturing companies is not based solely on equipment control capability. Manufacturing companies must also be efficient at coordinating and controlling personnel, materials, and equipment across different control systems in order to reach their maximum potential. This is usually accomplished using software systems and documented procedures that are collectively called the "manufacturing execution system (MES)" layer. MES defines a diverse set of functions that operate above automation control systems, reside below the level of enterprise business systems, and are local to a site or area. This chapter explains the functions of the MES layer and how these functions integrate with other corporate business systems.

The ANSI/ISA-95.00.03-2005 - *Enterprise-Control System Integration, Part 3: Models of Manufacturing Operations Management* standard defines 5 levels of activities in a manufacturing organization. Automation and control supports one level and MES supports a higher level, as shown in Figure 25-1.

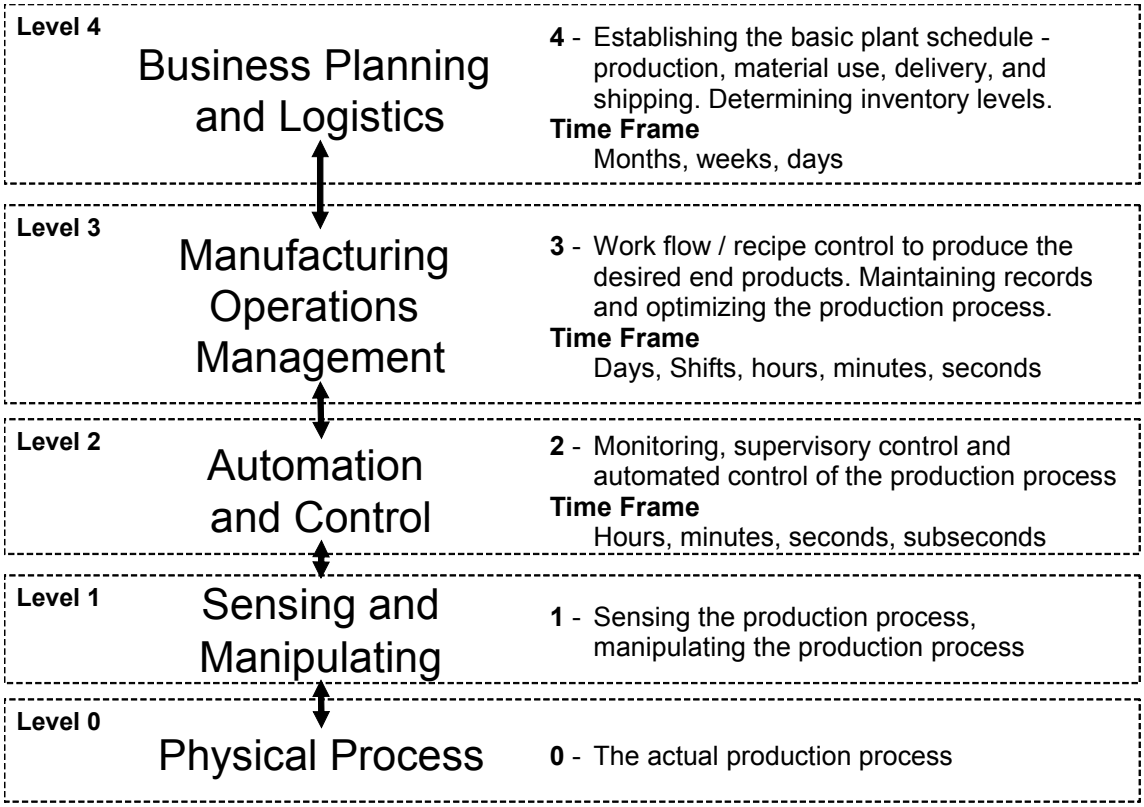


Figure 25-1: Activity Hierarchy in a Manufacturing Company

- ***Level 0*** defines the actual physical processes.
- ***Level 1*** defines the activities involved in sensing and manipulating the physical processes. Level 1 elements are the sensors and actuators attached to the control functions in automation systems.
- ***Level 2*** defines the activities of monitoring and controlling the physical processes and in automated systems this includes equipment control and equipment monitoring. Level 2 automation and control systems have real-time responses measured in subseconds and are typically implemented on programmable logic controllers (PLC), distributed control systems (DCS), and open control systems (OCS).
- ***Level 3*** defines the activities that coordinate production resources to produce the desired end products. It includes, work-flow "control" and procedural "control" through recipe execution. Level 3 typically operates on time frames of days, shifts, hours, minutes, and seconds. Level 3 functions also include maintenance functions, quality assurance and laboratory functions, and inventory movement functions, and are collectively called Manufacturing Operations Management. Level 3 functions directly related to production are usually automated using manufacturing execution systems (MES).
- ***Level 4*** defines business-related activities that manage a manufacturing organization. Manufacturing-related activities include establishing the basic plant schedule (such as material use, delivery, and shipping), determining inventory levels, logistics "control," and material inventory "control" (making sure materials are delivered on time to the right place for production). Level 4 is called Business Planning and Logistics. Level 4 typically operates on time

frames of months, weeks, and days. Enterprise resource planning (ERP) logistics systems are used to automate Level 4 functions.

It is important to remember that each level has some form of control and each level has its own definition for real-time. Level 3 systems consider *real-time* to mean information available a few seconds after shop floor events occur. Level 4 systems consider *real-time* to mean logistics and material information is available daily or within a few hours after the end of a shift.

25.2 MES Integration with Business Planning and Logistics

The ANSI/ISA-95.00.01-2000 - *Enterprise-Control System Integration Part 1: Models and Terminology* and ANSI/ISA-95.00.02-2001 - *Enterprise-Control System Integration Part 2: Object Model Attributes* standards define terminology to be used for interfaces between Level 3 systems and Level 4 systems. This information is used to direct production activities and to report on production.

Formal data models for exchanged information include:

Personnel Class, Person, and Qualification Test Information – This is the definition of the persons and personnel classes (roles) involved in production. This information may be used to associate production with specific persons as part of a production record, or with personnel classes to allocate production costs.

Equipment Class, Equipment, and Capability Test Information – This is the definition of the equipment and equipment classes involved in production. This information may be used to associate production with specific equipment as part of a production record, or with equipment classes to schedule production and allocate costs.

Material Class, Material Definition, Material Lot, Material Sublot, and QA Test Information – This is the definition of the lots, sublots, material definitions, and material classes involved in production. This information allows Level 3 and Level 4 systems to unambiguously identify material specified in production schedules and consumed or produced in actual production.

Process Segment Information – This is the definition of the business views of production, based on Level 4 business processes that must send information to production, or receive information from production. Examples include: setup segments, inspection segments, production segments, and cleanup segments.

Product Definition Information – This is the definition of the materials, equipment, personnel, and instructions it takes to make a product. This includes the *Manufacturing Bill* (a subset of the Bill of Material [BOM] that contains the quantity and type of material required for producing a product). It also includes product segments, which define the routing and specific resources required at each segment of production.

Production Capability Information – This is the definition of the capability and capacities available from production for current and future periods of time. Capability and capacity information is required for both Level 4 scheduling and Level 3 detailed production scheduling.

Production Schedule Information – This specifies what products are to be made. It may include the definition of the specific personnel or roles to be used, equipment or equipment classes to be used, material lots or material classes to be produced, and material lots or material classes to be consumed for each segment of production.

Production Performance Information – This specifies what was actually produced. It may include the definition of the actual personnel or personnel classes used, the actual equipment or equipment classes used, the actual material lots and quantities consumed, and the actual material lots and quantities produced for each segment of production.

25.3 Level 3 Equipment Hierarchy

Figure 25-2 shows the equipment and organizational hierarchy defined in the ANSI/ISA-95.00.03-2005 - *Enterprise-Control System Integration, Part 3: Models of Manufacturing Operations Management* standard. Level 4 ERP Logistics systems will typically coordinate and manage the entire enterprise and sites within the enterprise, but it may also schedule to the area or work center level. Level 3 MES systems will typically coordinate and schedule areas, work centers, and work units.

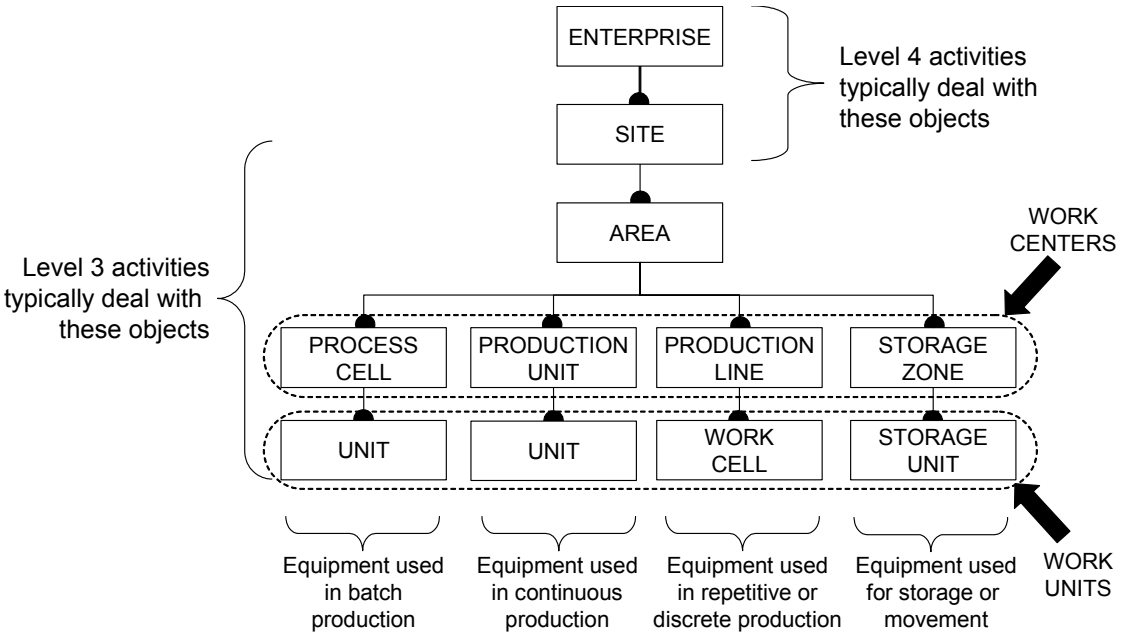


Figure 25-2: Equipment Hierarchy for Level 3 and Level 4 Functions

The equipment hierarchy is an expansion of the equipment hierarchy defined in the ANSI/ISA-88.01-1995 batch control standard to include equipment types used in continuous production, discrete production, and inventory storage and movement. The equipment hierarchy provides a standard naming convention for the organization of equipment, automation control, and manual control.

25.4 MES and Production Operations Management

Figure 25-3 illustrates the different Level 3 production-oriented functions that take place in sites and areas. Each bubble in the figure represents a collection of activities that occur in a production facility as a production schedule is converted into actual production. It illustrates how production requirements from the business are used to coordinate and control plant floor activity. The top four arrows identify previously defined information that is exchanged with business logistics systems.

The production model is driven by *production schedules* developed by the business and sent to production. The *production schedules* are used by detailed production scheduling activities that define *detailed production schedules* containing *production work orders*. The *production work orders* are dispatched to work centers and work units based on time and events, the *production work order* is executed and data is collected in a production data collection activity. (**Note**: In batch systems a *control recipe* is the equivalent of a *production work order*.)

The collected data is used in production tracking activities that relate the time-series information to the work order information to generate a report on *production performance* and tracing and tracking

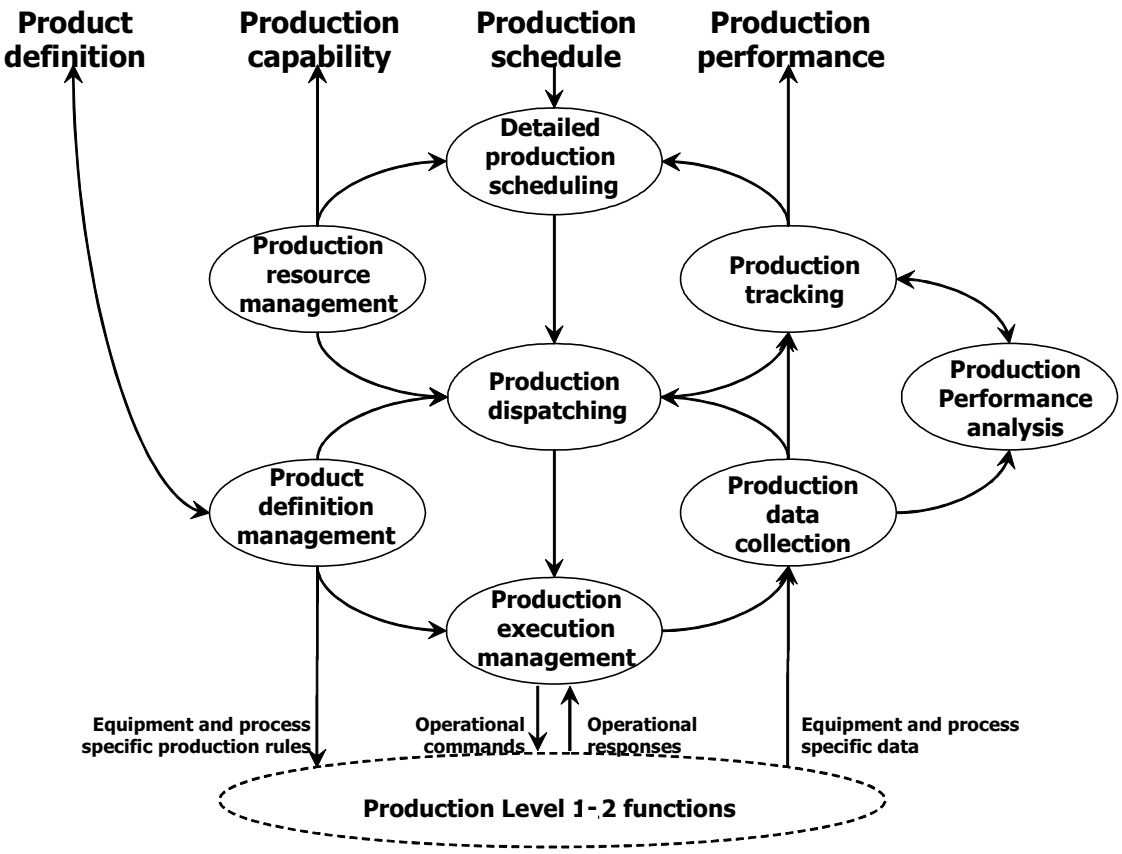


Figure 25-3: Manufacturing Operations Management Functions

information. The collected data and the data from tracing and tracking is used in production analysis functions to generate reports and KPIs (Key Performance Indicators). *Production capability* information about the current and future availability is provided to business scheduling systems by production resource management activities. *Product definition* information about the recipe, procedures, Bill of Material (BOM), and work routing needed for production is managed by product definition management activities.

25.5 Detailed Production Scheduling

These are the activities in a facility that take a production schedule and use information about local resources to generate a detailed production schedule. This can be an automated process, but in many plants scheduling is done manually by expert production planners or production planning staff. Automated systems are sometime referred to as plant level advanced planning and optimization systems. The key element of this activity is detailed scheduling of work assignments and material flows to a finer level of granularity than the business schedule. While Level 4 schedules may schedule work assignments to areas and work centers, detailed production scheduling will schedule work assignments to work centers and work units.

25.5.1 Production Dispatching

Once a detailed production schedule is available, that schedule is dispatched to production lines, process cells, production units, and storage zones. This can take the form of supervisors receiving daily schedules and dispatching work to technicians, or automated systems performing campaign management of batches and production runs. Production dispatching includes handling conditions not anticipated in the detailed production schedule. This may involve judgment in managing workflow and

buffers. Unanticipated conditions may have to be communicated to maintenance operations management, quality operations management, and/or inventory operations management. This is one of the core functions of an MES.

25.5.2 Production Execution Management

Production execution management activities receive the dispatched work requests and, using paper based systems, MES systems, or recipe execution systems, coordinate and control the actual work execution. This may include the execution of procedural logic in recipes and display of work flow instructions to operators. The activities include selecting, starting, and moving units of work (such as a batch or production run) through the appropriate sequence of operations to physically produce the product. The actual equipment control is part of the Level 2 functions. Production execution management is one of the core functions of an MES system, but it may also be performed by recipe or manual work flow instruction systems in DCS systems or batch execution systems. The standards for information flows from Level 3 to Level 2 are defined in the ANSI/ISA-88.01-1995, OPC, and Fieldbus standards.

25.5.3 Production Data Collection

Production data collection are the activities that gather, compile, and manage production data for specific units of work (batches or production runs). Manufacturing control systems generally deal with process information such as quantities (weight, units, etc.), properties (rates, temperatures, etc.), and equipment information such as controller, sensor, and actuator statuses. Collected production data includes sensor readings, equipment states, event data, operator-entered data, transaction data, operator actions, messages, calculation results from models, and other data of importance in the making of a product. The collected data is inherently time or event based, with time or event data added to give context to the collected information. This information is usually made available to various analysis activities, including product analysis, production analysis, and process analysis. Real-time data historians and automated batch record logging systems support this activity.

25.5.4 Production Tracking

The production tracking activities convert sensor and equipment data into information related to assigned work (batches and production runs), and into tracking information about equipment, material, and personnel used in production. Production tracking also merges and summarizes information that is reported back to the business activities. This is one of the core functions of an MES. When automated systems are used they usually link to data historians and batch record logging systems.

25.5.5 Production Resource Management

The resource management activities monitor the availability of personnel, material, and equipment production resources. This information is used by detailed production scheduling and business logistics planning. These activities take into account the current and future predicted availability, using information such as planned maintenance and vacation schedules, in addition to material order status and delivery dates. This activity may also include material reordering functions, such as Kanban. Kanban is a material management system used as part of just-in-time production operations where components and sub-assemblies are produced, based upon notification of demand from a subsequent operation. A Japanese word for “sign,” Kanbans are typically a re-order card or other method of triggering new production of material based on actual usage.

Resource management is usually a mixed operation, with manual work, automation, and database management. Management of the resources may include local resource reservation systems, and there may be separate reservation systems for each type of managed resource (personnel, equipment, and material). This is one of the core functions of an MES.

25.5.6 Product Definition Management

Product definition management includes activities associated with the management of product definitions. These may be recipes, work instructions, assembly instructions, standard operating procedures,

and other information used by production to make or assemble products. This is one of the core functions of an MES.

25.5.7 Production Performance Analysis

The activities associated with the analysis of production, process, and product are defined as production performance analysis. These are usually off-line activities that look for ways to improve processes through chemical or physical simulation, analysis of good and bad production runs, and analysis of delays and bottlenecks in production. Production performance analysis also includes calculating performance indicators, leading, and trailing predictors of behavior. These activities generally are major users of information collected in plant data historians. There are often separate tools for production, process, and product analysis, and the tool sets vary based on the type of production (continuous, discrete, or batch).

25.6 Other Manufacturing Activities

The above list does not define all of the activities of a production facility. There are also maintenance operations management activities, quality operations management activities, and inventory operations management activities.

Maintenance Operations Management – The activities that coordinate, direct, and track the functions that maintain the equipment, tools and related assets to ensure their availability for manufacturing.

Quality Operations Management – The activities that coordinate, direct, and track the functions that measure and report on quality. The broad scope of quality operations management includes both quality operations and the management of those operations to ensure the quality of intermediate and final products.

Inventory Operations Management – The activities that coordinate, direct, and track the functions that transfer of materials between and within work centers and manage information about material locations and statuses.

Manufacturing operations also require infrastructure activities that may be specific to manufacturing, but which are often elements also required by other parts of a manufacturing company. The infrastructure activities include:

a) Managing security within manufacturing operations

b) Managing information within manufacturing operations

c) Managing configurations within manufacturing operations

d) Managing documents within manufacturing operations

e) Managing regulatory compliance within manufacturing operations

f) Managing incidents and deviations

25.7 Level 3-4 Boundary

There are four rules which can be applied to determine if an activity should be managed as part of Level 4 or as part of Levels 3, 2, or 1. An activity should be managed at a Level 3 or below if the activity is directly involved in manufacturing, includes information about personnel, equipment, or material, and meets any of the following conditions:

a) The activity is critical to plant safety.

b) The activity is critical to plant reliability.

c) The activity is critical to plant efficiency.

d) The activity is critical to product quality.

e) The activity is critical to maintaining product or environmental regulatory compliance.

Note: This includes such factors as safety, environmental, and cGMP (current good manufacturing practices) compliance.

This means, in some cases, the Level 3 activities defined above may be performed as part of logistics instead of operations. Typically, this involves detailed production scheduling and production dispatching. The scope of an MES system is determined by applying the above rules to each site or area within a site.

25.8 References

1. ANSI/ISA-95.00.01-2000. *Enterprise/Control System Integration – Part 1: Models and Terminology.*

2. ANSI/ISA-95.00.02-2000. *Enterprise/Control System Integration – Part 2: Object Model Attributes*.

3. ANSI/ISA-95.00.03-2005. *Enterprise/Control System Integration – Part 3: Models of Manufacturing Operations Management*.

4. Cox III, James F. and John H. Blackstone, Jr. *APICS Dictionary*. Ninth Edition. APICS, 1998.

5. *MES Functionalities and MRP to MES Data Flow Possibilities*. White Paper Number 2. MESA International, 1994.

6. Williams, Theodore J. *The Purdue Enterprise Reference Architecture: A Technical Guide for CIM Planning and Implementation*. ISA, 1992.

25.8.1 Practical References

1. Brown, Mark Graham. *Baldrige Award Winning Quality: How to Interpret the Malcolm Baldrige Award Criteria.* ASQC Quality Press, 1995 and 1996.

2. Goldratt, Eliyahu M. and Jeff Cox. *The Goal: A Process of Ongoing Improvement.* North River Press, 1992.

About the Author

Dennis Brandl is the chief consultant for BR&L Consulting, specializing in Manufacturing IT applications, including business-to-manufacturing integration, MES solutions, general, and site recipe implementations, and automation system security. He has been involved in automation system design and implementation in a wide range of applications over the past 25 years. They have included biotech, pharmaceutical, chemical plants and oil refineries, food manufacturing, consumer packaged goods, PLC-based systems, and batch control systems. He is an active member of ISA's SP95 Enterprise/Control System Integration Committee and is editor of the set of SP95 standards. He has a BS in Physics and an MS in Measurement and Control from Carnegie-Mellon University, and an MS in Computer Science from California State University.